



Predicting the Fermi Level of Transition Metal-Doped Iron Surfaces Using Machine Learning.

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Presentation Outline



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Introduction

- ❖ Global warming is a pressing global issue that has drawn serious attention from the scientific community.
- ❖ CO₂ is the primary greenhouse gas driving climate change.
- ❖ Iron-based materials are vital in catalysis, CO₂ conversion, and energy applications, supporting climate change mitigation.
- ❖ A key property governing their performance is the Fermi level, which can be significantly tuned by doping iron surfaces with transition metals ^[7]

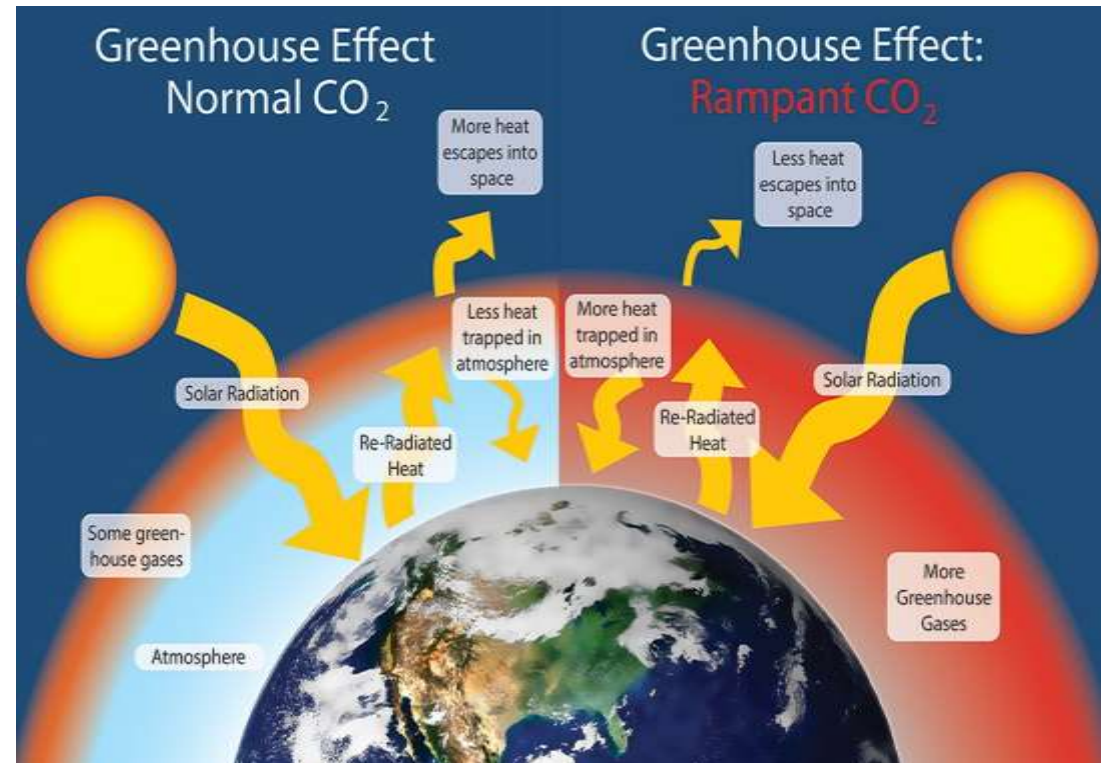


Fig 1. A schematic of greenhouse effect

Credit: Will Elder, National park service

Introduction



- ❖ The Fermi level of a metal is the energy of the highest occupied electronic state at a given temperature.
- ❖ It can be viewed as the electronic chemical potential, indicating how easily electrons transfer between a catalyst surface and adsorbed molecules such as CO_2 .
- ❖ In materials science and chemistry, the Fermi level is a fundamental property for designing advanced materials in catalysis, energy storage, and electronic devices.

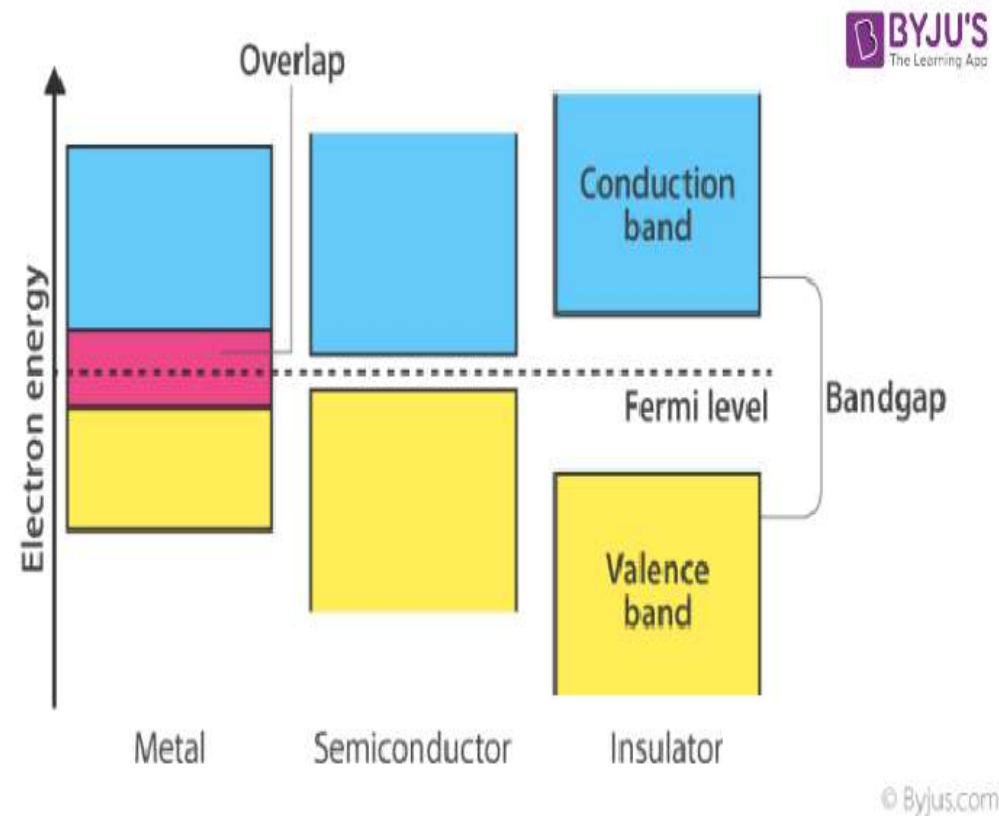


Fig 2. Fermi level position for different materials

Literature Review



- ❖ Tang et al. (2024) demonstrated that transition-metal dopants (Co, Ni, Cu, Zn) in Fe catalysts shift the d-band center and Fermi level, tuning intermediate adsorption and steering CO₂ hydrogenation activity and product selectivity.
- ❖ Roy et al. (2023), developed ML models (neural network and random forest) to predict the work function of MXenes with different surface terminations; changes in termination electronegativity shift the Fermi level and tune electronic properties.
- ❖ Takigawa et al. (2016) utilized machine learning models to predict d-band centers of metals and alloys from simple descriptors (density, enthalpy of fusion), enabling rapid screening of catalytic materials.

Problem Statement



- ❖ The catalytic performance of transition-metal-doped iron surfaces for CO₂ conversion is governed by their Fermi level, but calculating it with DFT is costly. A fast, predictive approach is needed to accelerate catalyst discovery.

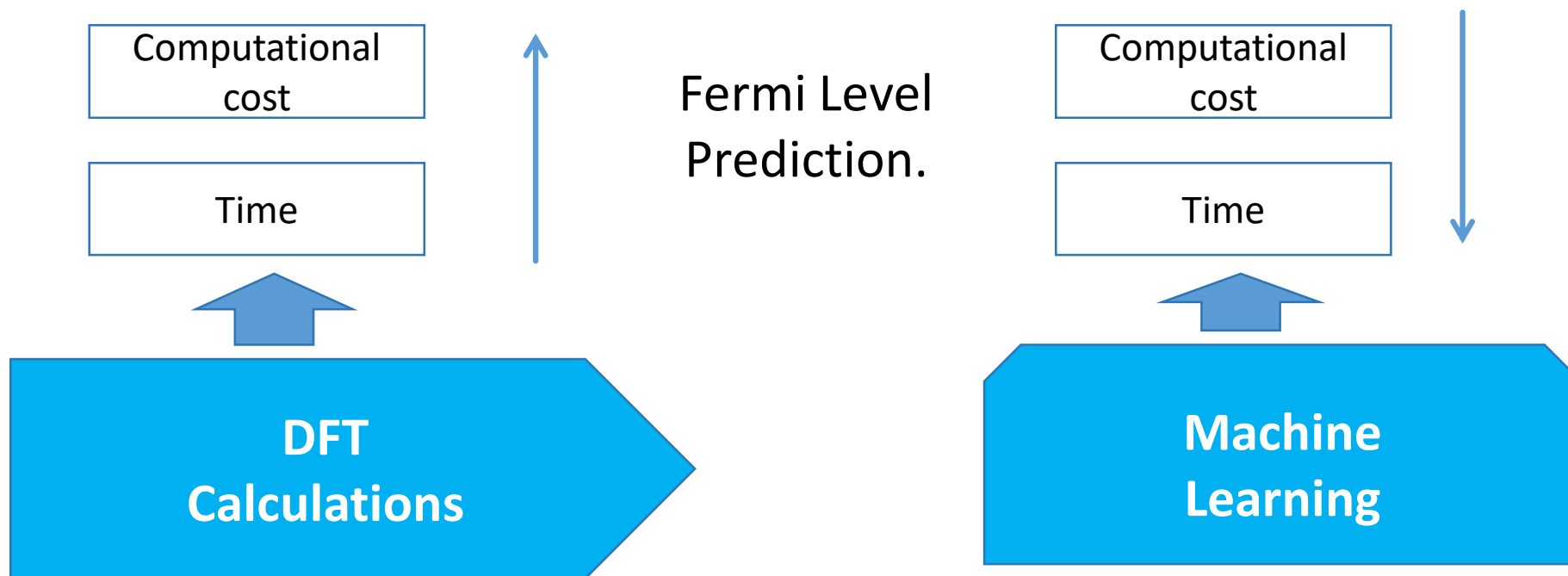


Figure 3

Justification



- ❖ Machine learning enables rapid optimization of materials by learning directly from pre-calculated properties.
- ❖ By leveraging structural and electronic descriptors, ML models can capture the underlying patterns that determine the Fermi level.
- ❖ This approach provides a fast and scalable alternative to traditional quantum mechanical calculations, accelerating the discovery of efficient catalysts.

Aims and Objectives



Aim

- ❖ To develop accurate and reliable AI models capable of predicting the Fermi level of transition metal–doped iron surfaces.

Objectives

- ❖ To model doped iron structures using Python and ASE.
- ❖ To perform DFT calculations on transition metal–doped iron surfaces and extract their Fermi levels.
- ❖ To establish relationships between chemical/compositional descriptors and DFT-computed Fermi levels using supervised machine learning.

Computational Details and Method

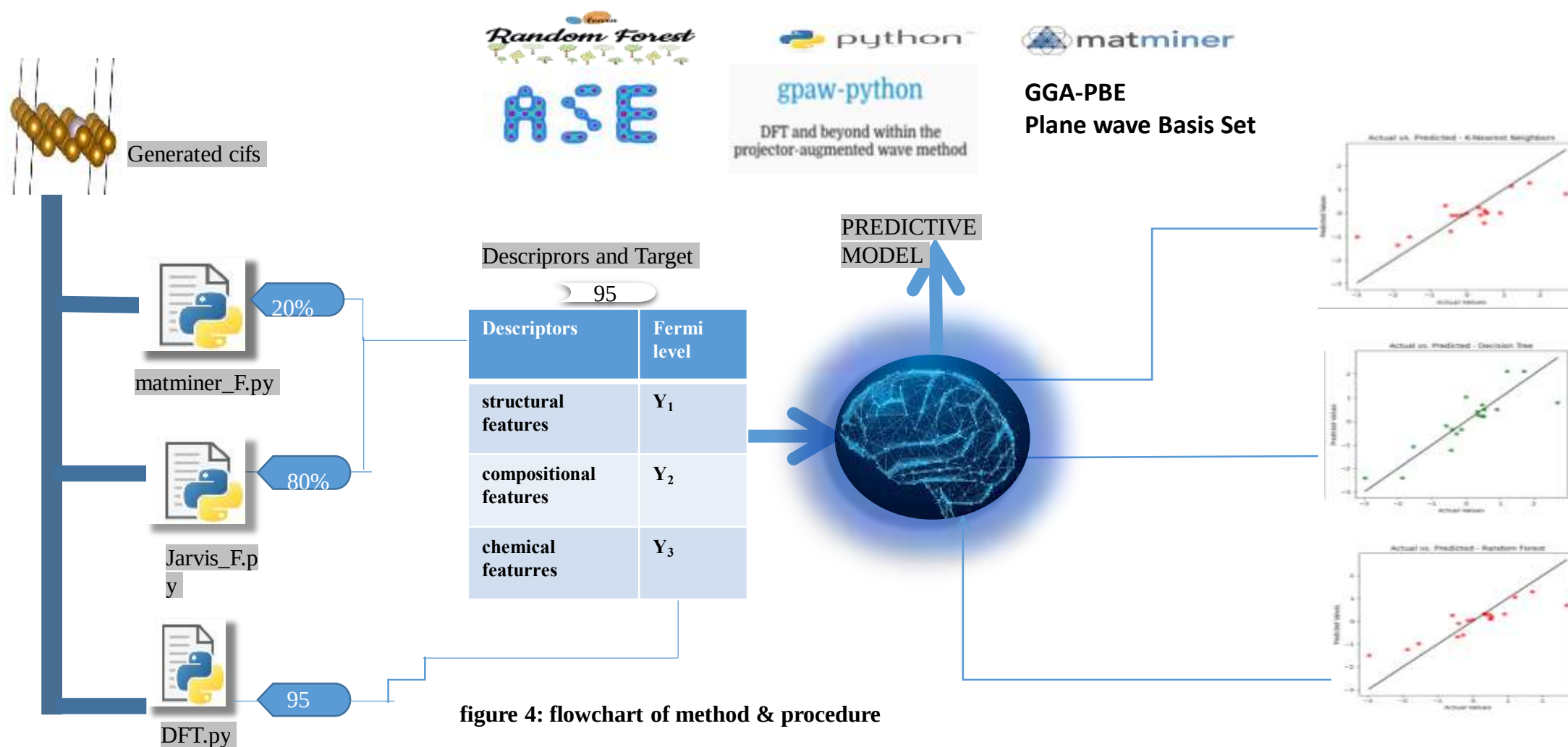
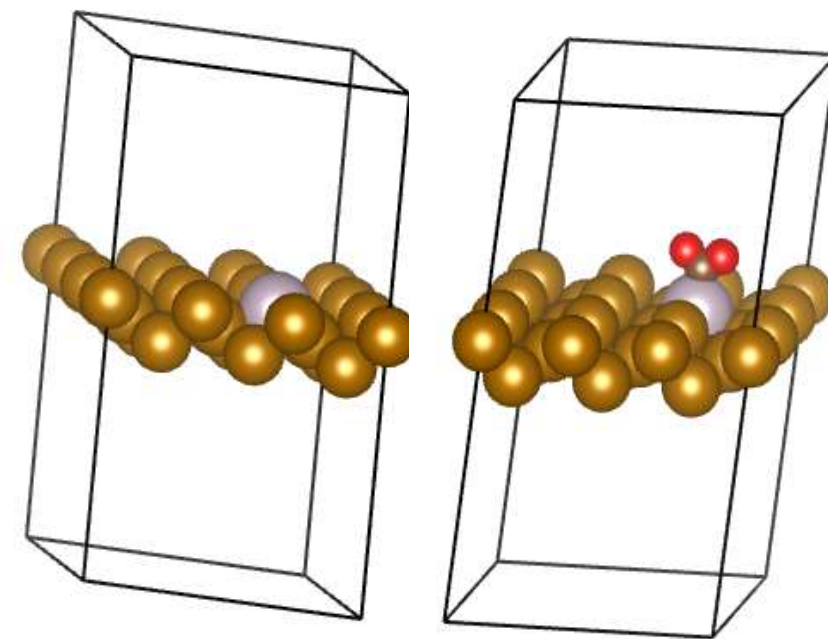


figure 4: flowchart of method & procedure

Results And Discussion

- ❖ A total of 96 transition-metal-doped iron structures were generated.
- ❖ Doping was performed for 32 transition metals, each substituted on the iron surface.
- ❖ Three Fe-bcc surface orientations were considered: (100), (110), and (111).



doped iron slab

doped slab + CO₂

figure 5

Results And Discussion



- ❖ A total of 96 structures were calculated, yielding 96 target values. After data cleaning, 95 targets were retained for model training.

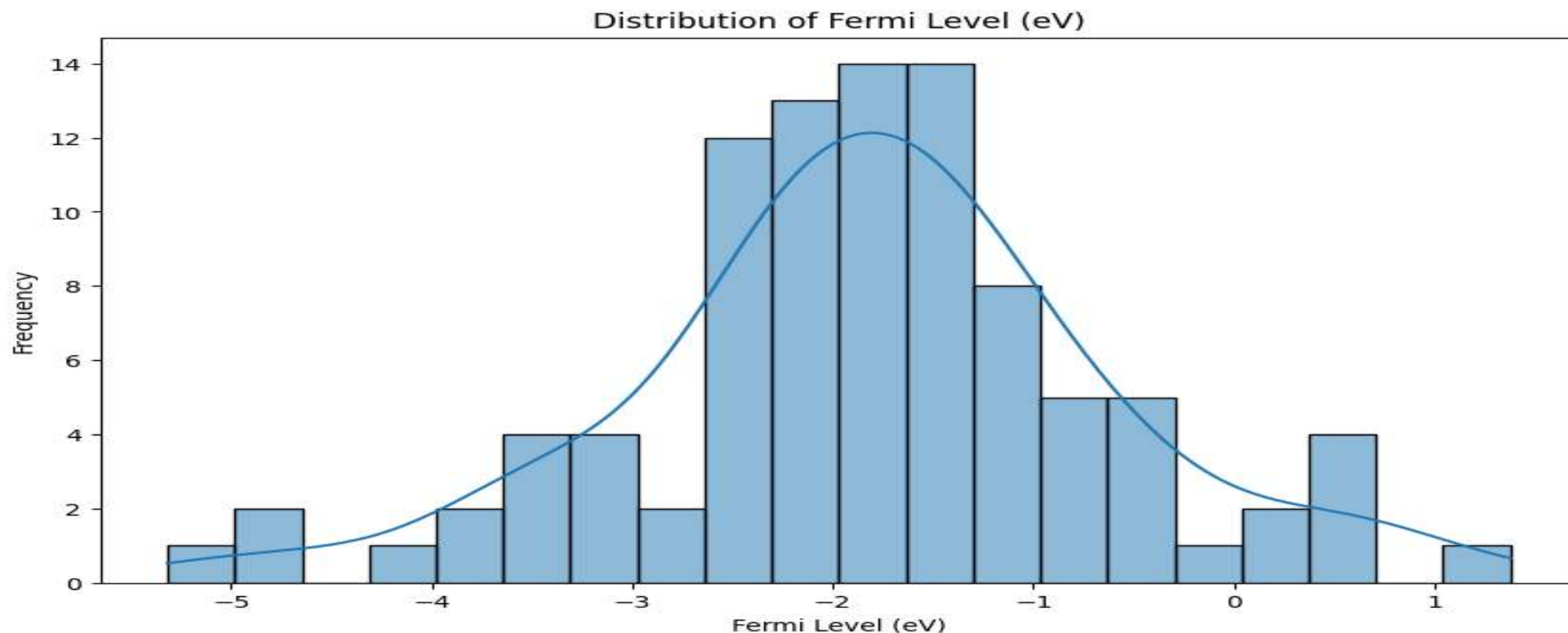


Figure 6. Calculated DFT data obtained

Results And Discussion



- ❖ Using Matminer and JARVIS-ML featurizers, chemical, structural, and compositional descriptors were obtained for the doped iron structures.
- ❖ Matminer (Materials Data Mining Library): an open-source Python library for extracting descriptors from materials data, widely used in materials informatics.
- ❖ JARVIS-ML (Joint Automated Repository for Various Integrated Simulations – Machine Learning): a toolkit providing pre-computed materials descriptors and ML utilities for accelerated materials discovery.
- ❖ In total, 87 descriptors were generated.
- ❖ After feature selection, 7 key descriptors (combining chemical and structural features) were used to train the ML model.

Results And Discussion

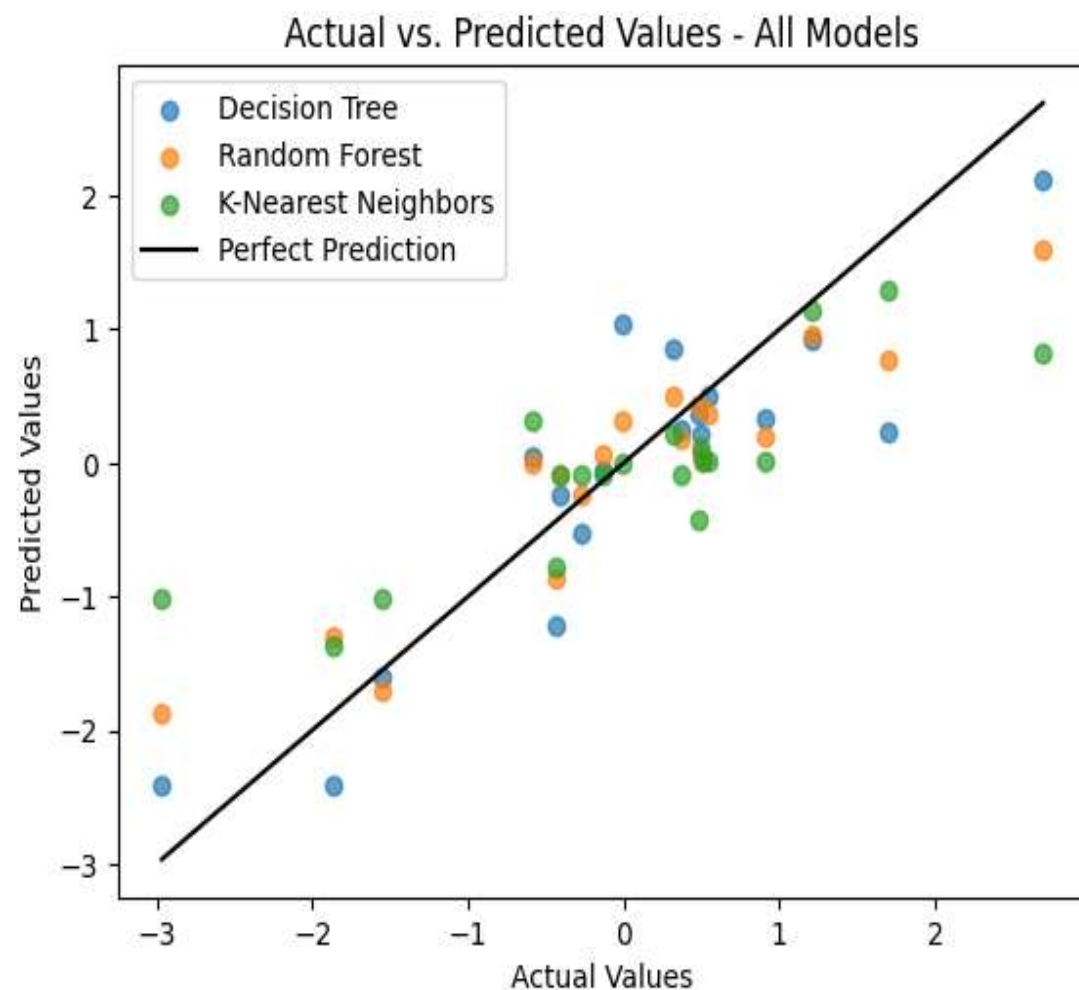


Figure 7a. Actual vs predicted plot

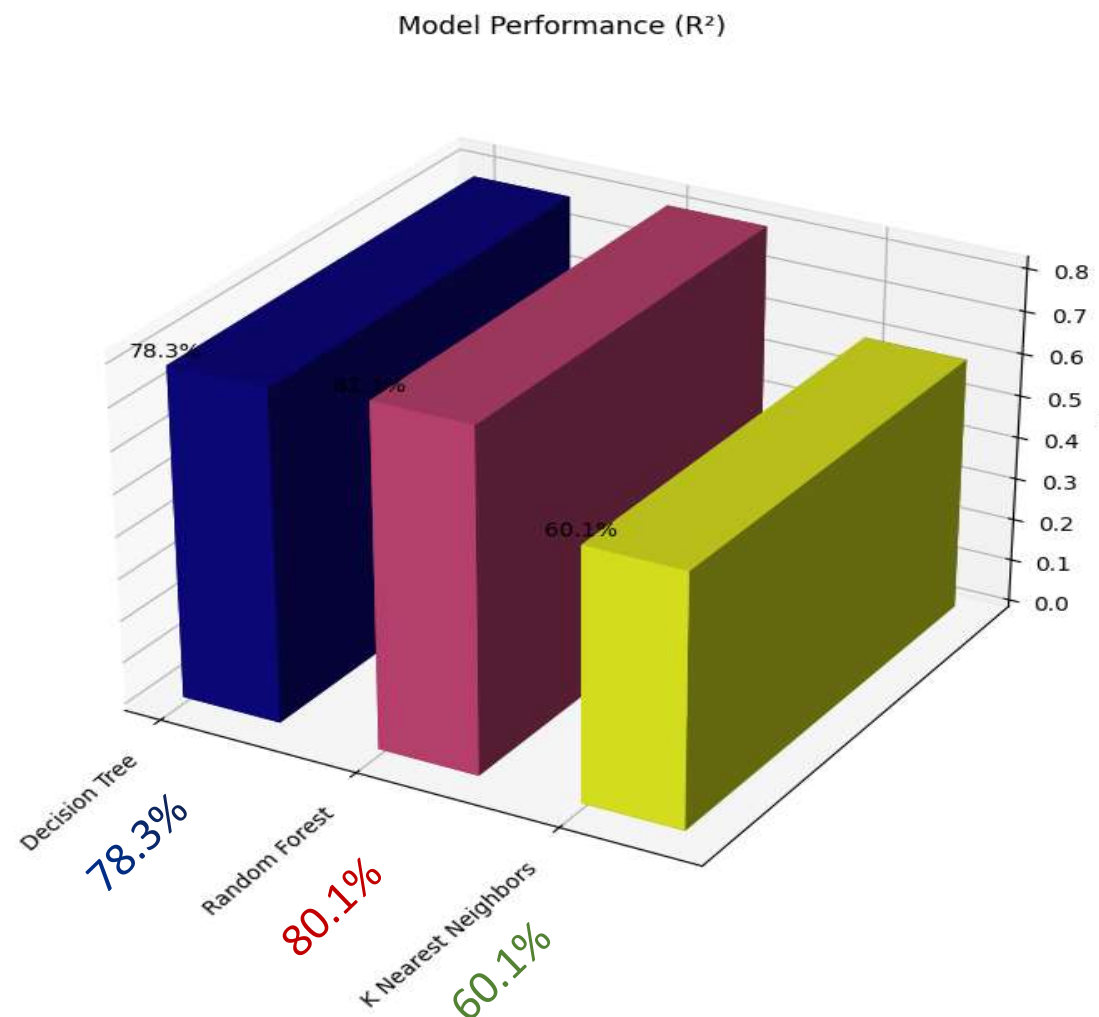


Figure 7b. Model performance chart

Results And Discussion

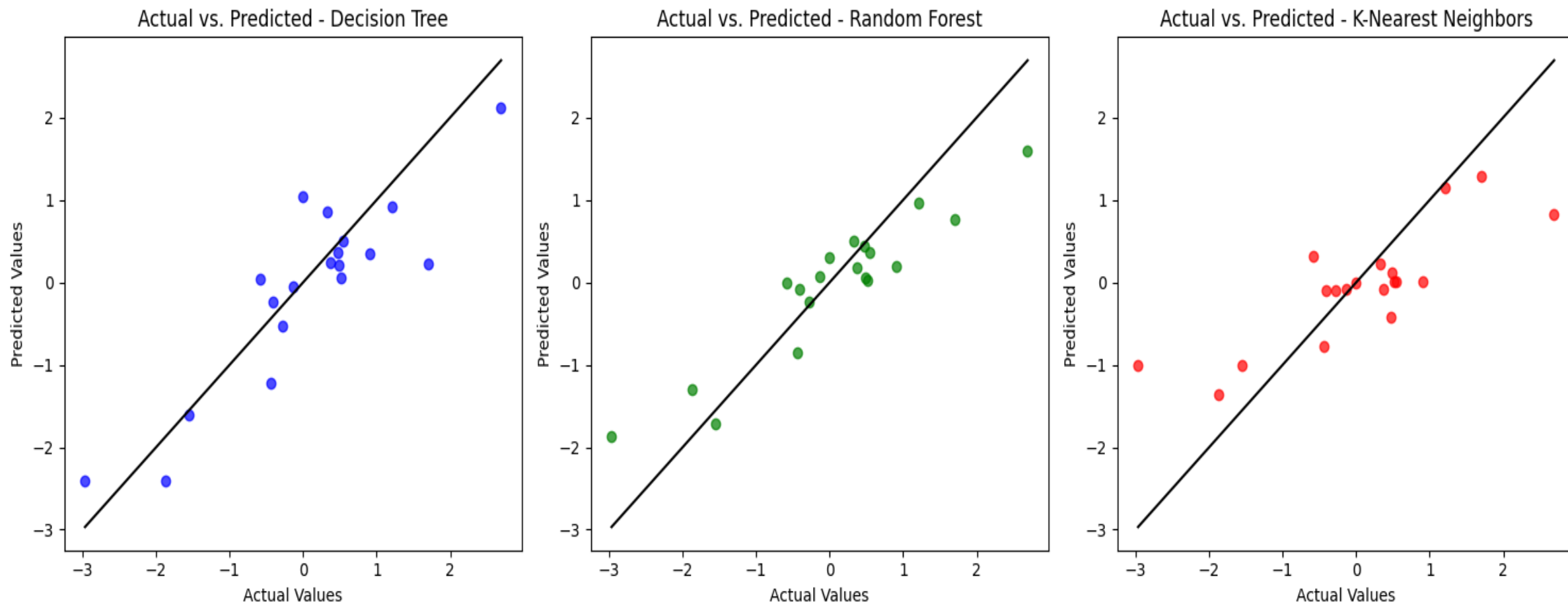


Figure 8. Actual vs predicted for each of the three models

Results And Discussion

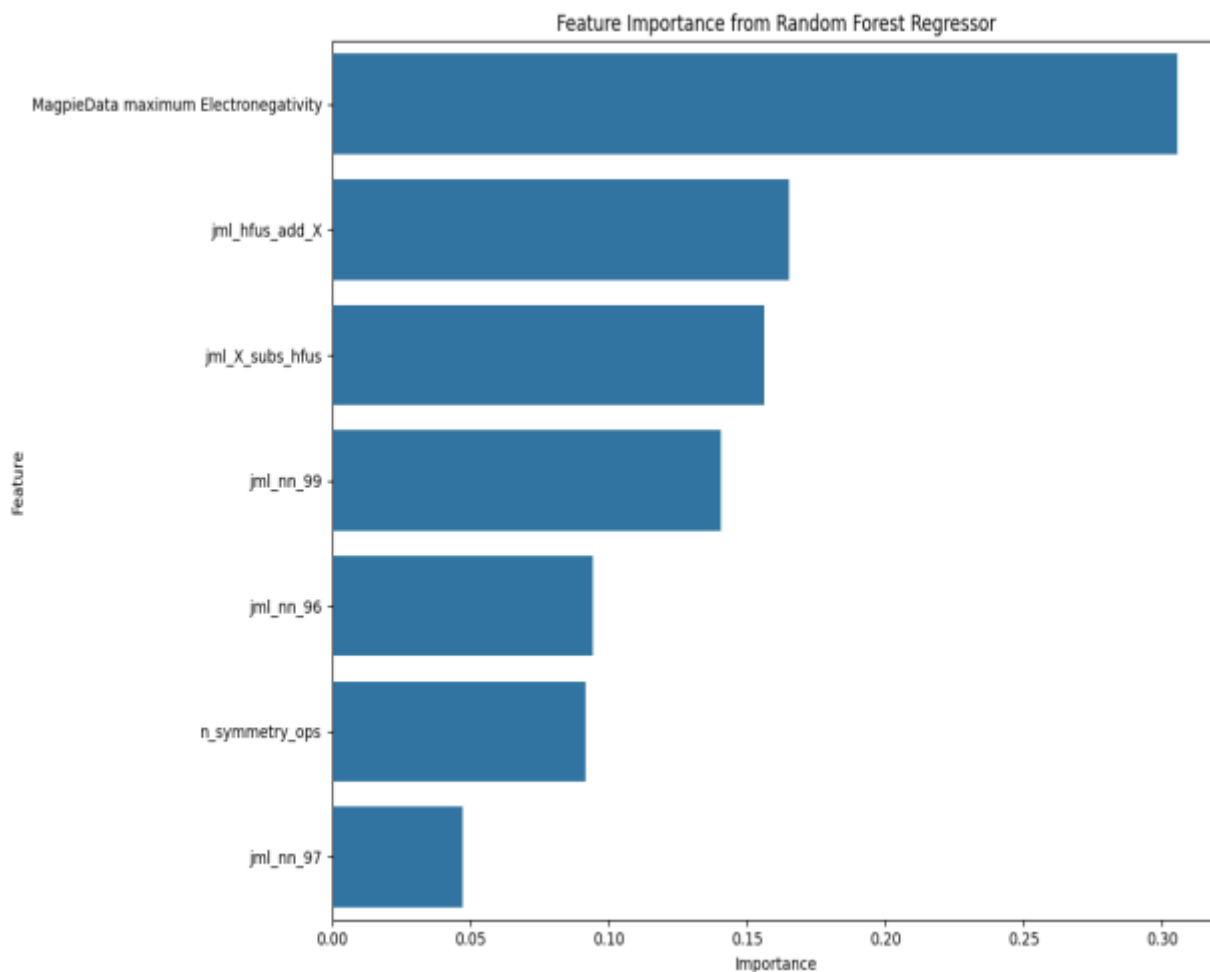


Figure 9a. Feature importance graph

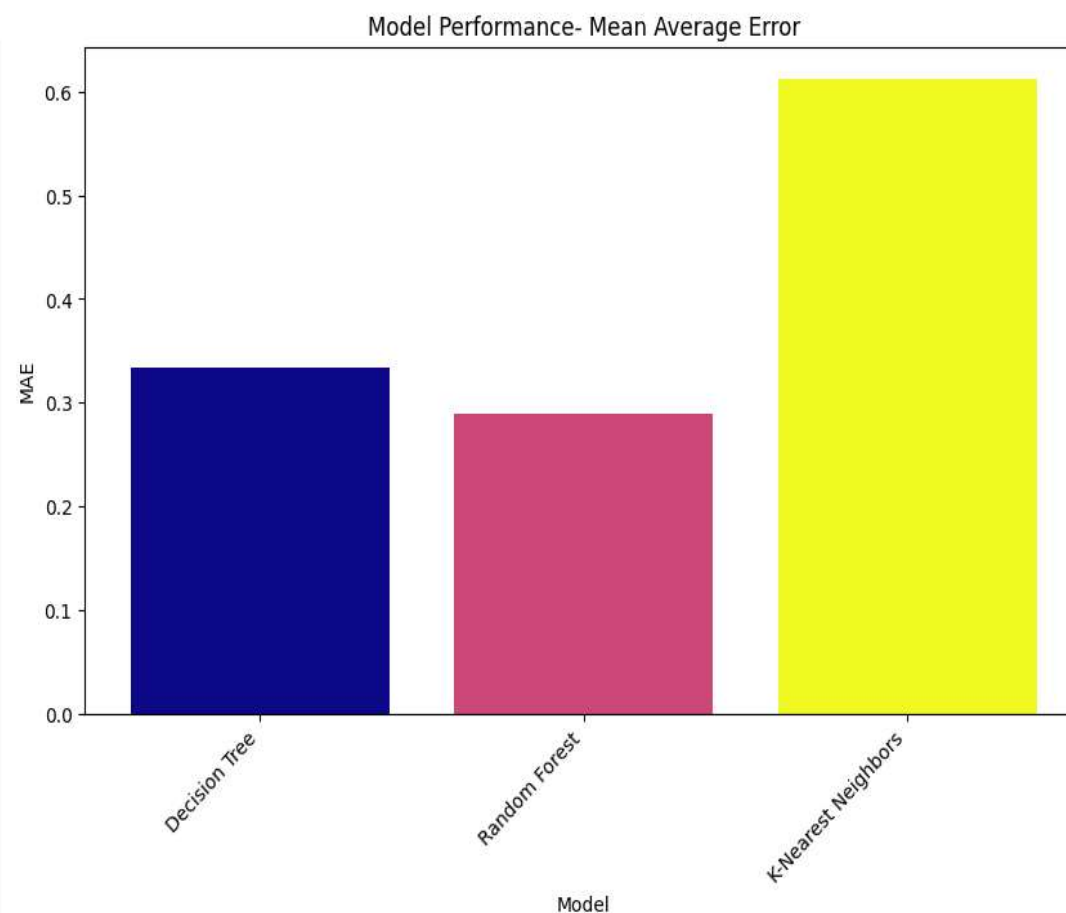


Figure 9b. Bar chart of the MAE of the three models

Results And Discussion



Material	DFT	Decision trees	RFR	K-Nearest neighbor
Fe_{26}Ni	-1.756 eV	-1.398 eV	-1.559 eV	-0.911 eV
SrFe_2O_3	1.56 eV	1.211 eV	1.358 eV	1.037 eV
Percentage Deviation				
Fe_{26}Ni		20.39%	11.22%	48.12%
SrFe_2O_3		22.37%	12.95%	33.31%

Conclusion



- ❖ Machine learning models were successfully developed to predict the Fermi level of transition metal–doped iron surfaces using DFT data and compositional descriptors.
- ❖ The Random Forest model achieved the best performance, explaining ~81% of the variance with a low prediction error (≈ 0.2893), showing that AI can effectively replicate DFT results for faster materials design.
- ❖ Key descriptors influencing the predictions included maximum electronegativity, `jml_hfus_add_X`, `jml_X_subs_hfus`, and `jml_nn_99`, highlighting the importance of both electronic and structural features.

THANK YOU
THANK YOU

References



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2. Herbig A. et al. (2015) investigated the impact of various transition metal substitutions in iron-pnictide superconductors. Their first-principles study revealed that such substitutions introduce mobile charge carriers and cause significant shifts in the Fermi level, thereby altering the electronic structure of the host material.
3. Takigawa et al. (2016) utilized machine learning to predict the d-band centers for eleven metals and their bimetallic alloys. By employing readily available descriptors such as density and enthalpy of fusion, they developed models that accurately estimated d-band centers, facilitating the rapid screening of catalytic materials.
4. Zhang P. et al. (2014) conducted an in-situ surface doping study on $\text{FeTe}_{0.55}\text{Se}_{0.45}$. They observed that electron doping via potassium evaporation shifts the Fermi level, allowing access to previously unoccupied electronic states and modifying the material's electronic properties.
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